

Nested Tuple

A tuple within another tuple is called as nested tuple or nesting of a tuple.

Ex:-

a = (10, 20, 30, (50, 60))

b = (50, 60)

a = (10, 20, 30, b)

a = ((10, 20, 30), (40, 50, 60))

a = ((10, 20, 30),
 (40, 50, 60))

Index

$a = (10, 20, 30, (50, 60))$

$b = (50, 60)$

$a = (10, 20, 30, b)$

10	20	30	<table><tr><td>50</td><td>60</td></tr><tr><td>[0]</td><td>[1]</td></tr></table>	50	60	[0]	[1]
50	60						
[0]	[1]						
[0]	[1]	[2]	[3]				

$a = ((10, 20, 30), (40, 50, 60))$

$a = ((10, 20, 30),$
 $(40, 50, 60))$

	0	1	2
0	10	20	30
1	40	50	60

Accessing Nested Tuple

```
a = (10, 20, 30, (50, 60))
```

```
b = (50, 60)
```

```
a = (10, 20, 30, b)
```

```
print(a[0])
```

```
print(a[1])
```

```
print(a[2])
```

```
print(a[3][0])
```

```
print(a[3][1])
```

10	20	30	<table><tr><td>50</td><td>60</td></tr><tr><td>[0]</td><td>[1]</td></tr></table>	50	60	[0]	[1]
50	60						
[0]	[1]						
[0]	[1]	[2]	[3]				

```
print(a)
```

All elements

Accessing Nested Tuple

```
a = ( (10, 20, 30),  
      (40, 50, 60) )
```

```
print(a[0][0])
```

```
print(a[0][1])
```

```
print(a[0][2])
```

```
print(a[1][0])
```

```
print(a[1][1])
```

```
print(a[1][2])
```

```
print(a)
```

All elements

	0	1	2
0	10	20	30
1	40	50	60

Accessing Nested tuple using for loop

```
a = ( (10, 20, 30),  
      (40, 50, 60) )
```

Without index

```
for r in a:  
    for c in r:  
        print(c)  
    print( )
```

Outer for loop

Inner for loop

Inner for loop

With index

```
n = len(a)  
for i in range(n):  
    for j in range(len(a[i])):  
        print(a[i][j])  
    print( )
```

Outer for loop

The outer for loop represents the rows and the inner for loop represents the columns in each row.

Accessing Nested tuple using while loop

```
a = ( (10, 20, 30),  
      (40, 50, 60) )
```

```
n = len(a)
```

```
i = 0
```

```
while i < n :
```

```
    j = 0
```

```
    while j < len(a[i]):
```

```
        print(a[i][j])
```

```
        j+=1
```

```
    i+=1
```